# Ian Waudby-Smith

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## Academic positions

**University of California, Berkeley** — **Berkeley, CA**
*Miller Postdoctoral Fellow in Statistics* — *2024–2027*
Host: Michael I. Jordan

## Education

**Carnegie Mellon University** — **Pittsburgh, PA**
*PhD, Statistics* — *2019–2024*
Advisor: Aaditya Ramdas
Umesh K. Gavaskar Best Dissertation Award

**University of Waterloo** — **Waterloo, Ontario**
*BMath, Pure Mathematics & Statistics* — *2013–2018*
5-year co-op program
Dean's Honours List

## Papers

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Gaffke's confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *arXiv preprint arXiv:2607.18661*, 2026.

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Combining e-values via demi-supermartingales. *arXiv preprint arXiv:2603.10329*, 2026.

Ricardo J. Sandoval⋆, **Ian Waudby-Smith**⋆, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925*, 2026.

Ben Chugg⋆, Etienne Gauthier⋆, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *arXiv preprint*, 2026.

Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments. *arXiv preprint*, 2026.

**Ian Waudby-Smith**, Ricardo J. Sandoval, and Michael I. Jordan. Universal log-optimality for general classes of e-processes and sequential hypothesis tests. *arXiv preprint*, 2025+.

**Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *arXiv preprint*, 2025+.

**Ian Waudby-Smith**, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform anytime-valid sequential inference. *arXiv preprint*, 2023+.

($\alpha\beta$) Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *Bulletin of the London Mathematical Society* (*accepted*), 2026.

**Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability* (*accepted*), 2026.

**Ian Waudby-Smith**, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.

**Ian Waudby-Smith**, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/JMS Journal of Data Science* (***Invited contribution***), 1(3):1–42, 2024.

**Ian Waudby-Smith** and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology* (***Discussion paper***), 86(1):1–27, 2024.

**Ian Waudby-Smith**, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning* (***Oral presentation***), 2023.

Akash V. Maharaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.

**Ian Waudby-Smith**, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting* (***Best paper award***), pages 124–139. Springer, 2021.

**Ian Waudby-Smith** and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems* (***Spotlight***), 33:20204–20214, 2020.

**Ian Waudby-Smith**, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.

**Ian Waudby-Smith**, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

($\alpha\beta$) Alphabetical ordering.
*Equal contribution.

## Teaching Experience

**Carnegie Mellon University** — **Pittsburgh, PA**
*Graduate Teaching Assistant* — *2019–22*
- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

## Other experience

**Google Research** — **New York, NY**
*Student Researcher* — *Jun–Aug 2023*
Mentors: Jean Pouget-Abadie & Jennifer Brennan

**Microsoft Research** — **New York, NY & Redmond, WA**
*Research Intern* — *May–Aug 2022*
Mentor: Paul Mineiro

**Adobe Research** — **San Jose, CA**
*Research Intern* — *Jun–Aug 2020*
Mentors: David Arbour & Ritwik Sinha

**The Hospital for Sick Children (SickKids)** — **Toronto, Ontario**
*Research Student* — *Apr–Aug 2019*
Mentor: Eleanor Pullenayegum

**Health Data Science Lab, University of Waterloo** — **Waterloo, Ontario**
*Research Assistant* — *2016–18*
Mentors: Joel Dubin & Joon Lee

**Department of Statistics, University of Waterloo** — **Waterloo, Ontario**
*Research Assistant* — *Apr–Aug 2017*
Mentor: Pengfei Li

**Department of Pure Mathematics, University of Waterloo** — **Waterloo, Ontario**
*Research Assistant* — *Jan - Apr 2017*
Mentor: Yu-Ru Liu

**Cancer Care Ontario** — **Toronto, Ontario**
*Student Analyst* — *Jan–Apr 2016*

Mentor: Zhihui (Amy) Liu

## Service

**Thesis committee member:** Tyron Lardy (PhD in Statistics, Leiden University), Noah Liniger (MSc in Data Science, ETH Zurich)

**Reviewer:** The Annals of Statistics, Journal of the Royal Statistical Society (Series B), Bernoulli, Biometrika, The Electronic Journal of Probability, The Electronic Journal of Statistics, The Journal of the American Statistical Association, STOC, The New England Journal of Data Science, Sankhya A.

**University of California, Berkeley** — **Berkeley, CA**
*Volunteer*
- Miller symposium organizing committee
- Incoming Miller fellow initiative

**Carnegie Mellon University** — **Pittsburgh, PA**
*Volunteer*
- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

## Awards

**Institute of Mathematical Statistics** — **Berkeley, CA**
*New Researcher Travel Award* — *2026*

**Miller Institute for Basic Research in Science** — **Berkeley, CA**
*Miller fellowship* — *2024–'27*

**Carnegie Mellon University Department of Statistics and Data Science** — **Pittsburgh, PA**
*Umesh K. Gavaskar Best Dissertation Award* — *2024*

**Statistical Society of Canada** — **St. John's, NL**
*Probability Section Student Research Presentation Award* — *2024*

**Amazon Science** — **Pittsburgh, PA**
*Graduate Research Fellowship* — *2023*

**University of Waterloo** — **Waterloo, Ontario**
*Waterloo Statistics Student Conference Presentation Award* — *2022*

**Carnegie Mellon University Department of Statistics and Data Science** — **Pittsburgh, PA**
*Teaching Assistant of the Year* — *2021*

**Adobe Research** — **Pittsburgh, PA**
*PhD Research Gift* — *2020*

**University of Waterloo** — **Waterloo, Ontario**
*David Johnston International Experience Award* — *2018*

**The Natural Sciences and Engineering Research Council of Canada** — **Waterloo, Ontario**
*NSERC Undergraduate Student Research Award* — *2017*

**University of Waterloo** — **Waterloo, Ontario**
*President's Research Award* — *2016–17*

**University of Waterloo** — **Waterloo, Ontario**
*University of Waterloo President's Scholarship of Distinction* — *2014*

# Presentations

## Short courses

**MBZUAI** — **Abu Dhabi, UAE**
*A brief introduction to game-theoretic, safe, anytime-valid inference* — *2025*
Mini course consisting of 4 lectures

**ERC OCEAN retreat** — **Venice, Italy**
*A brief introduction to game-theoretic, safe, anytime-valid inference* — *2024*
Mini course consisting of 3 lectures

## Department seminars

**Department of Statistics at the London School of Economics** — **London, UK**
*Log-optimality and multi-armed sequential hypothesis testing* — *2026*

**Department of Statistics and Actuarial Science at the University of Waterloo** — **Waterloo, Ontario**
*Log-optimality and multi-armed sequential hypothesis testing* — *2026*

**Department of Statistics at University of California, Davis** — **Davis, CA**
*Log-optimality and multi-armed sequential hypothesis testing* — *2026*

**Department of Statistics at Iowa State University** — **Ames, IA**
*Anytime-valid inference and uniform central limit theory* — *2026*

**Division of Biostatistics at the University of California, Berkeley** — **Berkeley, CA**
*Anytime-valid off-policy inference for contextual bandits* — *2025*

**Department of Statistics at Stanford University** — **Stanford, CA**
*Anytime-valid inference and uniform central limit theory* — *2025*

**Statistics Seminar at the University of Copenhagen** — **Copenhagen, Denmark**
*Anytime-valid off-policy inference for contextual bandits* — *2023*

## Invited talks at conferences / workshops / research groups

**Institute for Operations Research and the Management Sciences (INFORMS)** — **San Francisco, CA**
*Multi-armed log-optimality and sequential testing* — *2026*

**Joint Statistical Meetings (JSM)** — **Boston, MA**
*Asymptotic anytime-valid inference* — *2026*

**Adaptive Experimentation Workshop at University College London (Gatsby)** — **London, UK**
*Log-optimality and multi-armed sequential hypothesis testing* — *2026*

**International Workshop on Sequential Methodologies** — **Washington, D.C.**
*Asymptotic anytime-valid statistical inference* — *2026*
Recipient of the IMS New Researcher Travel Award

**Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis)** — **Stanford, CA**
*On nonasymptotic confidence intervals for treatment effects in randomized experiments* — *2026*

**National University of Singapore Young Mathematical Scientists Forum** — **Singapore**
*Log-optimality of e-processes and sequential tests* — *2025*

**Stanford Data Driven Seminar (host: Stefan Wager)** — **Stanford, CA**
*Log-optimality of e-processes and sequential tests* — *2025*

**MBZUAI-Berkeley Workshop** — **Abu Dhabi, UAE**
*Anytime-valid off-policy inference for contextual bandits* — *2025*

**Inria/Sierra Seminar** — **Paris, France**
*Anytime-valid inference and uniform central limit theory* — *2025*

**International Seminar on Selective Inference** — **Virtual**
*$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X* — *2024*

**CLIMB Workshop** — **Berkeley, CA**
*Election audits via anytime-valid inference* — *2024*

**Workshop on Game-Theoretic Statistical Inference** — **Oberwolfach, Germany**
*$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X* — *2024*

**Fienberg Student Research Workshop at Carnegie Mellon University** — **Pittsburgh, PA**
*Election audits via anytime-valid inference* — *2024*

**Joint Statistical Meetings** — **Toronto, Ontario**
*Anytime-valid off-policy inference for contextual bandits* — *2023*

**International Conference on Machine Learning** — **Honolulu, HI**
*Extensions of randomized response for private confidence sets* — *2023*

**Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen)** — **Amsterdam, Netherlands**
*Anytime-valid off-policy inference for contextual bandits* — *2023*

**University of Toronto / The Vector Institute (host: Dan Roy)** — **Toronto, Ontario**
*Anytime-valid off-policy inference for contextual bandits* — *2023*

**Copenhagen Causality Lab, University of Copenhagen** — **Virtual**
*Asymptotic confidence sequences for anytime-valid causal inference* — *2023*

**Conference on Digital Experimentation (CODE@MIT)** — **Cambridge, MA**
*Asymptotic confidence sequences for anytime-valid causal inference* — *2022*

**Microsoft Research Reinforcement Learning Discussion Group** — **Virtual**
*Anytime-valid contextual bandit inference* — *2022*

**CMS, California Institute of Technology (host: Houman Owhadi)** — **Virtual**
*A brief introduction to safe, anytime-valid inference (SAVI)* — *2022*

**Microsoft Research** — **Virtual**
*A brief introduction to safe, anytime-valid inference (SAVI)* — *2022*

**Safe, Anytime-Valid Inference (SAVI) Workshop** — **Eindhoven, Netherlands**
*Time-uniform central limit theory and anytime-valid causal inference* — *2022*

**Carnegie Mellon University Computer Science Theory Lunch** — **Pittsburgh, PA**
*Estimating means of bounded random variables by betting* — *2021*

**International Seminar on Distribution-Free Statistics** — **Virtual**
*Estimating means of bounded random variables by betting* — *2021*

**E-Vote-ID: The International Conference for Electronic Voting** — **Virtual**
*RiLACS: Risk-limiting audits via confidence sequences* — *2021*

**Spotify Experimentation Platform Team** — **Virtual**
*Doubly robust confidence sequences for sequential causal inference* — *2021*

**Vinted Science and Analytics Meetup** — **Virtual**
*Doubly robust confidence sequences for sequential causal inference* — *2021*

## Other talks

**International Conference on Statistics and Data Science (ICSDS)** — **Seville, Spain**
*Concentration inequalities for strong laws of large numbers* — *2025*

**Statistical Society of Canada meeting** — **St. John's, NL**
*Distribution-uniform strong laws of large numbers* — *2024*

Recipient of the Probability Section's Student Presentation Award

**International Conference on Statistics and Data Science (ICSDS)** **Lisbon, Portugal**
*Distribution-uniform anytime-valid inference* *2023*

**Waterloo Student Conference in Statistics, Actuarial Science, and Finance** **Waterloo, Ontario**
*Estimating means of bounded random variables by betting* *2022*
Recipient of the Student Presentation Award

**TPDP: Theory and Practice of Differential Privacy Workshop** **Baltimore, MD**
*Locally private nonparametric confidence intervals and sequences* *2022*

**Statistical Society of Canada (SSC) Annual Meeting** **Virtual**
*Time-uniform central limit theory and anytime-valid causal inference* *2022*

**NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making** **Virtual**
*Time-uniform central limit theory and anytime-valid causal inference* *2021*

**Joint Statistical Meetings (JSM)** **Virtual**
*Doubly robust confidence sequences for sequential causal inference* *2021*

**Joint Statistical Meetings (JSM)** **Virtual**
*Confidence sequences for sampling without replacement* *2020*

**Statistical Society of Canada (SSC) Annual Meeting** **St. Catherines, Ontario**
*Multi-state models for chronic kidney disease prevalence projections in Ontario* *2016*